Biomedical Signal Processing and Control 100 (2025) 106963

Contents lists available at ScienceDirect

Biomedical Signal Processing and Control

journal homepage: www.elsevier.com/locate/bspc

An image enhancement based method for improving rPPG extraction under low-light illumination

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ARTICLE INFO

Keywords:
Deep-learning
Remote photoplethysmography
Heart rate estimation
Transfer learning
And image enhancement

ABSTRACT

Remote photoplethysmography (rPPG) has gained significant attention as a non-invasive approach for measuring human vital signs from videos. However, the reliance on capturing the reflection of ambient lighting causes it to be susceptible to the adverse effects of low illumination levels, leading to deterioration in the signal quality. The preliminary study introduces a novel methodology for enhancing rPPG signal extraction in low-light conditions through the integration of an Image Enhancement Model (IEM) inspired by Retinex theory, which significantly improved signal quality by preprocessing video frames to better capture the subtle changes in facial blood flow. Recognizing the challenge posed by the generalization capacity over different deep-learning models and unseen examples from various datasets, this study further evaluated the efficacy of the IEM + rPPG extraction model across multiple datasets (UBFC-rPPG, BH-rPPG, MMPD-rPPG, and VIPL-HR) by integrating IEM with existing deep-learning based rPPG extraction models including DeepPhys, PhysNet, and PhysFormer and traditional POS rPPG extraction method. Our experiments demonstrate a consistent improvement for all the methods in heart rate estimation accuracy across all datasets, underscoring the IEM's adaptability and effectiveness. The paper also explores the application of our method to traditional rPPG extraction techniques, further validating its potential for broader usage. Through comprehensive analysis, this research not only confirms the impact of lighting conditions on rPPG signal quality but also provides a robust solution for more reliable non-invasive vital sign monitoring in diverse environments.

1. Introduction

Photoplethysmography (PPG) was first used in developing survival suits to protect wearers against extreme external temperatures [1]. It is an optical technique that non-invasively measures blood volume changes in microvascular tissues. Remote photoplethysmography (rPPG) is an extended approach that uses light reflected from the facial regions to measure the vital signs related to blood flow, including heart rate (HR), heart rate variability (HRV), and blood pressure (BP). The technology has offered unprecedented convenience and safety in instances in which physical contact with the individual for vital sign measurements is unsuitable, such as cardiovascular disease management, neonatal monitoring, and intensive care units [2–4].

Despite its promise, the quality of vital signs' measurements based on rPPG has been hampered by the presence of motion artifacts, short exposure time, and insufficient lighting in which the camera struggles to capture facial details, leading to an increase in the HR estimation error as shown in Fig. 1. Yang et al. [5] has reported that the performance of both traditional and deep-learning based rPPG extraction models is reduced significantly for different lighting conditions.

Based on Shafer's dichromatic model [6], rPPG extraction algorithms are expected to perform poorly in a low illumination environment. Our preliminary work in [7] has suggested that the implementation of an image enhancement model (IEM) can facilitate the recovery of the specular and diffuse reflection under low illumination, leading to better accuracy in HR estimation with PhysNet [8]. The preliminary study shows the potential to generalize deep-learning based methods into low illumination level conditions with the aim of IEM. This paper reports the full results validating the generalizability of IEM exhaustively with the PhysNet [8], the DeepPhys [9], and the PhysFormer [10] models which are representative deep-learning models for rPPG extraction on multiple public datasets. A thorough validation of the benefit of IEM will connect the existing literature and lay a firm foundation for future image enhancing technology for

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https://doi.org/10.1016/j.bspc.2024.106963
Received 4 December 2023; Received in revised form 30 July 2024; Accepted 21 September 2024
Available online 29 September 2024

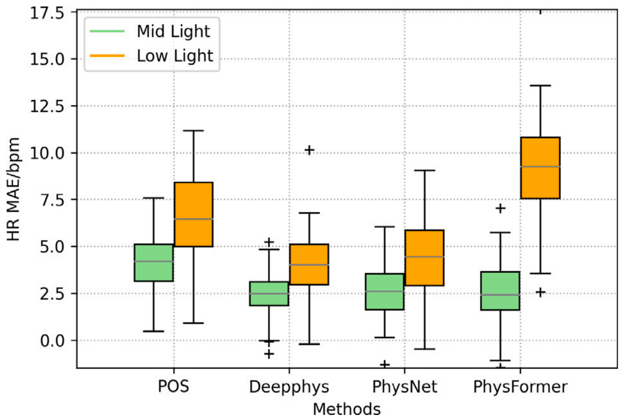

Fig. 1. HR MAE estimation error under different light conditions for POS, DeepPhys, PhysNet, and PhysFormer methods tested on BH-rPPG dataset under mid light and low light conditions.

rPPG extraction under low illumination conditions. The enhancement process can be conducted on a video basis, in which the frame rate or resolution is increased to improve the visibility of facial regions, or on an image basis, in which the sharpness, brightness, and contrast of the facial image are increased. Since the illumination intensity remains approximately consistent within each video clip [5,11], image enhancement is employed due to its computational efficiency.

The paper is organized as follows. Section 2 provides an overview of the historical context and progression of research in the field of rPPG extraction. Section 3 presents the design and framework of our innovative IEM + rPPG extraction model. The methodologies and outcomes of the experiments conducted to evaluate the effectiveness of our proposed model across various datasets and in comparison with other rPPG extraction models are elaborated in Sections 4 and 5, respectively. Section 6 offers analysis and interpretation of the findings reported in the preceding section. Finally, Section 7 concludes the study, summarizing its key insights and contributions.

2. Related work

2.1. Traditional methods for rPPG extraction

Extensive studies have been conducted on rPPG extraction, resulting in several traditional analytical methods, which consist of two categories: The first category is based on the fact that photoplethysmographic signals are highly dependent on both transmission and reflection of different wavelengths [12]. Martínez et al. [13] studied the optimum wavelength bands that carry useful information for decoding of PPG signal in both visible and NIR regions. Among other chrominance-based methods, POS [14] achieves state-of-the-art performance by eliminating specular reflections using a plane orthogonal to the skin tones in a temporally normalized RGB color space. The second category consists of statistical methods such as ICA-based and PCA-based methods [15,16]. ICA [15] hypothesizes that the RGB signal variation caused by blood flow is independent of other factors and separates independent signals from a linear mixture of sources based on a non-Gaussianity hypothesis. Similarly, the PCA method [16] assumes that the pulsatility of the PPG signal contributes to the periodic variation of RGB channels. By projecting the RGB channels into directions with the highest variation, PCA manages to identify the component that contains the most information about changes in blood flow by analyzing the principal components and selecting the one that corresponds to the rPPG signal.

However, these methods suffer in performance when the ambient light is inadequate. The physiological signals captured are inherently subtle and become even more challenging to discern in dim conditions, often being overshadowed by noise. Our preliminary results showed that the performance of traditional methods dropped significantly under low illumination conditions [7].

2.2. Deep-learning based rPPG extraction methods

Compared to traditional methods for rPPG-based HR estimation, which rely on handcrafted features, deep learning rPPG models employ a data-driven approach to directly learn the mapping between raw video frames and physiological signals. A 2D or 3D CNN supervised model is trained by pairing the raw videos with their corresponding PPG or Blood Volume Pulse (BVP) signals. [17] proposed an end-to-end HR estimation approach which is a two-step CNN that contains an extractor and HR estimator. The extractor is trained to maximize the signal-to-noise ratio (SNR) of rPPG signals from video frames and the output is fed into the estimator to predict HR value. The model performs better than traditional methods on different datasets.

DeepPhys [9] is a VGG 2D CNN that simultaneously trains a motion and appearance model. The motion model is constructed from the normalized difference between the adjacent video frames as its input. The appearance model, containing information about the tone and texture of the skin, then guides motion representation learning via an attention mechanism. By doing so, the network learns a spatial mask to detect the regions of interest (ROI) for predicting the rPPG signals. Additionally, the mechanism allows the spatial distribution of physiological signals to be visualized. A similar structure was adopted by [18] in which they employed a temporal shift module (TSM) that enables information exchange among neighboring frames without expensive 3D convolution operations.

Nevertheless, while 3D CNNs can capture the temporal context features of a video to obtain richer information contained in sequences of data, they are computationally costlier than 2D CNNs. PhysNet [8] is a 3D CNN that simultaneously utilizes spatial and temporal domains to derive rPPG features. In this CNN implementation, an encoder–decoder structure is implemented to reduce the noise and redundancy of the temporal domains. When tested on different input video lengths, this network shows superior performance to PhysNet-2DCNN+RNN with a shorter input duration, showing its superior ability to learn temporal representation. In [19], they utilized neural architecture search (NAS) to find the best-suited backbone 3D CNN for rPPG extraction. NAS was performed on two gradient-based NAS methods to form a backbone network for rPPG extraction.

Lu et al. [20] introduced Dual-GAN, simultaneously modeling both the BVP signal and noise distribution, leveraging two Generative Adversarial Networks (GANs) to improve the robustness and accuracy of physiological signal estimation. Specifically, the BVP-GAN focuses on learning a noise-resistant mapping to predict BVP signals accurately, while the Noise-GAN is dedicated to modeling the distribution of noise present in the videos. This synergistic approach allows for enhanced feature disentanglement between BVP signals and noise, leading to superior performance in HR estimation from facial videos across various challenging conditions.

Recently, with the success of transformer models [21], Yu et al. [10] introduced a transformer-based architecture for rPPG measurement from facial videos. This novel approach, named PhysFormer, adeptly leverages temporal difference transformers to capture both local and global spatiotemporal features, enhancing the quasi-periodic rPPG signal representation without the need for extensive preprocessing or pre-training on large-scale datasets. By incorporating a unique supervision strategy that combines label distribution learning with dynamic constraints in the frequency domain, PhysFormer achieves superior performance across various benchmark datasets.

Overall, deep learning models outperform traditional models under normal illumination conditions, but they underperform when models pre-trained on normal illumination are tested on low illumination video footage. We selected [9], PhysNet [8], and PhysFormer [10] as a representation of mainstream neural network structures for rPPG extraction to study the capabilities of our proposed method.

2.3. rPPG extraction in low illumination

Xi et al. [22] proposed to apply a low-light image enhancement (LIME) algorithm for rPPG extraction in a low-light environment. This algorithm considered the image enhancement as an optimization problem and solved it iteratively. This algorithm improved the HR estimation performance on a privately collected dataset, but it was only evaluated with traditional rPPG extraction algorithms. Lin et al. [23] have reported that even in a well-designed laboratory setting with stationary participants, low illumination can adversely affect the quality of rPPG signals extracted from all the mentioned models. To address this issue, Yang et al. [5] applied data augmentation and image enhancement in an unsupervised manner to investigate their effects on the performance of rPPG-based HR estimation of participants across various illumination conditions. However, the performance gains of these methods are limited, and some frame-wise image augmentation methods (e.g., Gamma Correction [24], Histogram Equalization [25]) can even disrupt the temporal consistency. Results of a pilot study demonstrate the potential benefit of an rPPG model to guide the transfer learning of an image enhancement model with low-light videos [7]. Preliminary results indicate that the model can significantly increase the accuracy of estimated HR under low and high illuminations. This paper reports the full results with validations against different representative rPPG extractors and multiple datasets.

3. Methods

3.1. Dichromatic model

At the pixel level, a facial image captured by a camera consists of two major components: (1) the shading component representing the true color of the face and (2) the highlight component representing additional effects due to illumination intensity; illumination angle (i); viewing angle (e); and the phase angle (g = i+e). Shaffer's Dichromatic Reflection Model [6] has been the pioneer in modeling this phenomenon. The model states that an image consists of two distinct types of reflections —interface (specular/the highlight component) reflection and body (diffuse/the shading component) reflection. Both types can be decomposed into a relative power spectral distribution and a geometric scale factor, as summarized in the following equation,
$L(\lambda,t) = m_i(t,i,e,g)c_i(\lambda) + m_b(t,i,e,g)c_b(\lambda)$

where m_i and m_b are geometric scale factors that depend only on geometry for interface (hence, i) and body (hence, b) reflections, respectively. $c_i(\lambda)$ and $c_b(\lambda)$ are relative spectral power distributions that depend on the wavelength only for interface and body reflections, respectively. Assuming the cameras and the faces are stationary, differences in illumination levels only influence the spectral power density, which is the amplitude $c_i(\lambda)$ and $c_b(\lambda)$ in the equation with little effects on the geometric factors. Therefore, our idea is to isolate the spectral power density from the received color image mixture $L(\lambda)$ to reduce the influence of illumination variation and improve the robustness under different illumination levels, especially for low-light levels. This process can be realized by image enhancement.

3.2. Deep-learning based image enhancement model

Numerous studies have examined the use of deep learning methods for the enhancement of general low-illumination images. The relevant literature review and benchmarking are detailed in the cited literature [26]. Many state-of-the-art image enhancement models are based on Retinex theory [27], in which the received color image can be decomposed into two factors as follows, $L(\lambda,t) = I(\lambda)\cdot R(t)$ where $R(t)$ is called reflectance, referring to the appearance of the object, and $I(\lambda)$ is called the illumination map, depicting the illumination level in the surrounding environment. Compared with the dichromatic model in rPPG extraction, we find out that the rPPG-related body diffuse signal is included in the reflectance components. The idea of image enhancement is in sync with the rPPG extraction under low illumination levels. Therefore, we proposed to combine the image enhancement model with the rPPG extraction model to eliminate the influence of low light levels.

3.3. Proposed method

The proposed IEM + rPPG extraction model is shown in Fig. 2. The input image sequences (video) are first processed by the image enhancement model (IEM) into a reflectance and illumination map, of which only the reflectance is fed into the cascaded rPPG extraction model to output the final rPPG value. Heart rate is estimated from the predicted rPPG signal by taking the frequency that corresponds to the amplitude peak in the power spectrum of the rPPG signal.

During the training process, we adopted a transfer learning strategy to fine-tune a pre-trained image enhancement model to make it able to decouple the rPPG-related signal from illumination intensity maps, while the parameters in the rPPG extraction model are also pre-trained on the rPPG dataset with normal light and frozen during the entire process. The reasons for this strategy are as follows:

1. Re-training of the rPPG extraction model with data under low-light conditions will inevitably deteriorate the performance of the original model in normal light conditions. Instead, we utilized the IEM to reduce the effect of the different illumination levels, in other words, to standardize the illumination level of the input into normal light where the follow-up rPPG extraction model works properly;
2. The training of IEM is data-consuming, and we do not have a sufficient number of rPPG datasets with a variety of illumination levels. However, if we apply an IEM module trained on other computer vision (CV) datasets without fine-tuning, the performance would be degraded [7]. This is because the target in the original CV task is different from the rPPG extraction. Correspondingly, the loss function is far from the rPPG extraction task. Therefore, we utilized transfer learning to fine-tune the IEM to enable it to extract rPPG-related signals.

We employed the lightweight image enhancement network recently proposed in the literature [28] as a basic model with slight modification. The detailed structure is shown in Fig. 3. To adapt the IEM module with the DeepPhys model which takes frame difference as input, we added a Differential layer before the output of IEM. This model provides us with an ideal starting point for evaluating the performance of general image enhancement modules in the specific task of enhancing low-light facial videos to extract rPPG signals. In the next section, we refer to this network simply as the Image Enhancement Model (IEM), which will be fine-tuned specifically for rPPG signal extraction.

3.4. rPPG-toolbox

To ensure the consistency and reproducibility of research findings, we have harnessed a powerful tool known as the rPPG-ToolBox, a publicly accessible GitHub repository as documented in [29]. The toolbox encompasses an array of functionalities, including data loaders tailored to different datasets, trainers tailored for diverse deep learning models, as well as scripts for rPPG extraction and evaluation. In the toolbox documentation [30], the authors provide a codebase designed to provide researchers with a tool that reproduces various models, datasets, and training processes for result replication and performance benchmarking.

Our optimization strategy focuses on a unified tuning of several key aspects of the training process, including the use of the same Adam optimizer and learning rate scheduler, as well as the application of the same second-order Butterworth filter (with cut-off frequencies of 0.75 and 2.5 Hz) for filtering out the predicted PPG waveforms. This uniformity adjustment not only helps to ensure that the proposed methods are compared under fair conditions but also simplifies the process of reproducing the study results.

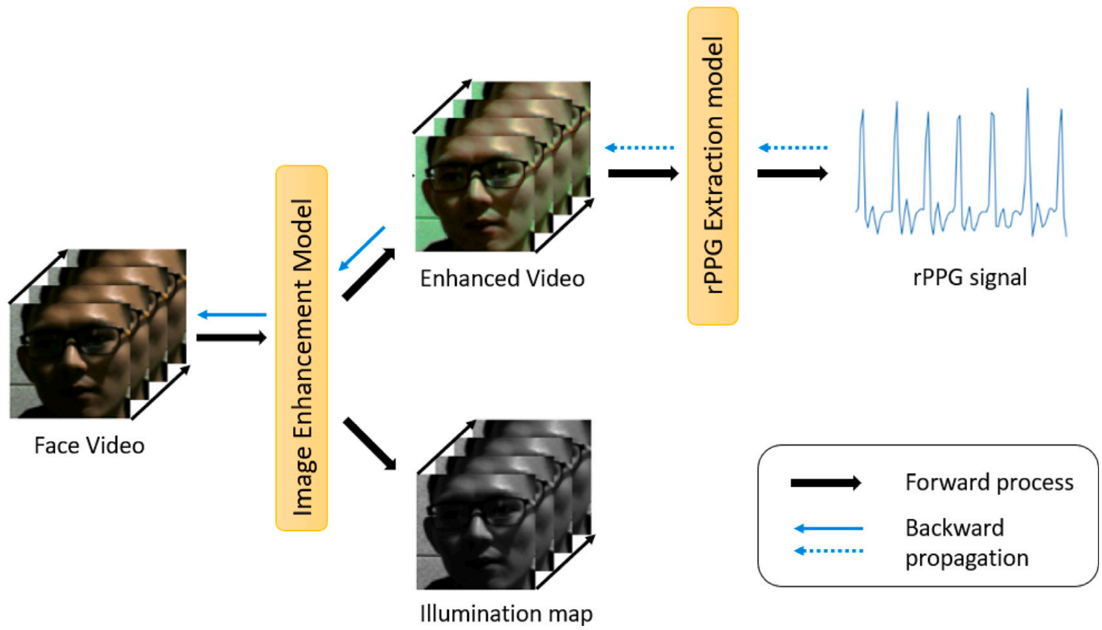

Fig. 2. Flowchart of the proposed IEM + rPPG extraction model. The black arrows show the forward process while the blue ones show backward propagation. The dashed and solid blue arrows correspond to the model parameters that are updated and not updated, respectively.
Source: Enhanced from [7].

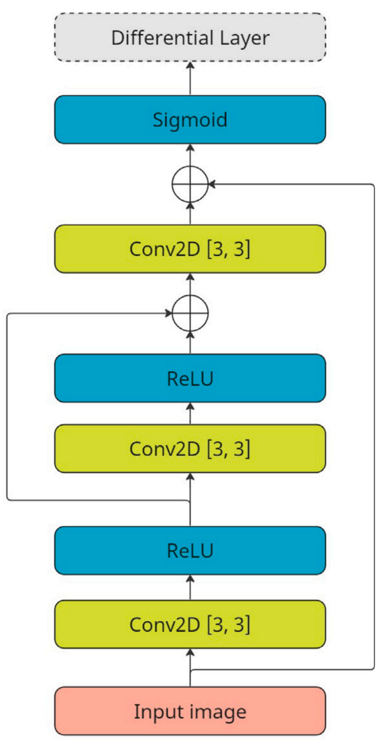

Fig. 3. Detailed structure of the image enhancement model (IEM). The differential layer is utilized to generate frame difference for the DeepPhys model and will be removed for other models using the raw frame as input.
Source: Enhanced from [7].

4. Experiments

4.1. Dataset

In our experiments, we used different datasets for model evaluation and transfer learning. Specifically, the UBFC-rPPG dataset [31] was used to generate the baseline rPPG extraction model because of the empirical fact that most rPPG extraction models show satisfying performance on the UBFC-rPPG dataset. The BH-rPPG dataset, MMPD-rPPG dataset, and VIPL-HR dataset were used to perform transfer learning and performance testing of rPPG extraction with an image enhancement model.

UBFC-rPPG dataset:

The UBFC-rPPG dataset [31] contains video recordings of 42 different subjects playing games in bright environments. The videos were recorded using a Logitech C920 HD Pro webcam at 30 frames per second with a resolution of 640 × 480 pixels in uncompressed 8-bit RGB format. To obtain ground truth data for PPG, they used a CMS50E transmission pulse oximeter.

BH-rPPG dataset:

The BH-rPPG dataset, described in detail in the literature [5], consists of 105 video recordings of 35 subjects under three different lighting conditions, low, medium, and high light, corresponding to light luminances of 8, 42.4, and 104 lux, respectively. The videos were recorded by using a Logitech HD pro-C310 color camera with a resolution set to 640 × 480 pixels and a frame rate of about 15 Hz. A CONTEC CMS50E transmissive pulse oximeter was employed to obtain ground truth data of the PPG signals.

MMPD-rPPG dataset:

The Multi-domain Mobile Video Physiology Dataset (MMPD) [11] was collected by Tsinghua University, which consists of 11 h of videos of 33 subjects recorded with mobile phones. The dataset contains a diversity of scenarios including different skin tones, body motions, lighting conditions, makeup, glasses wearing, and hair coverings. Two kinds of datasets were provided: full dataset with a resolution of 320 × 240 and mini dataset with a resolution of 80 × 60. The reference PPG data was recorded using an HKG-07C+ oximeter and downsampled from 200 Hz to 30 Hz to match the FPS of video data.

It should be noted that the MMPD-rPPG dataset contains more diversity of conditions for rPPG collection (i.e. skin tone, gender, makeup, glasses wearing, hair covering). For a fair comparison, we included all the different conditions. For the motion, we only selected the stationary trials as the intense motion would significantly degrade the rPPG extraction performance. For the exercise, we selected the trials

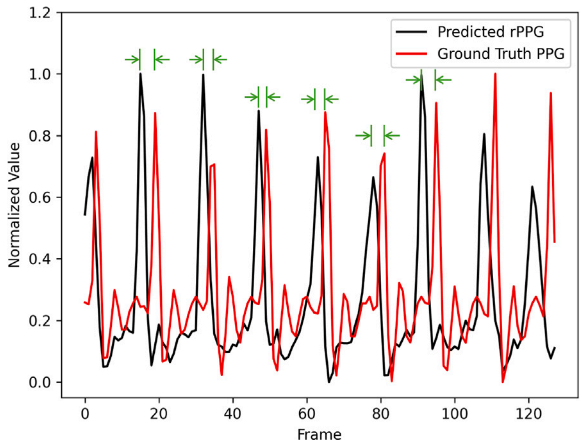

Fig. 4. Example of the misalignment between predicted rPPG and ground truth PPG extract by PhysNet with subject1 in BH dataset. The misalignments are marked with green arrows.

in which the subjects did not perform any exercise as exercise would also affect the rPPG extraction performance.

VIPL-HR dataset:

The VIPL-HR dataset [32] is designed to advance the field of remote heart rate (HR) estimation through facial video analysis under diverse and less restricted conditions. It comprises a comprehensive collection of 2,378 videos captured in visible light (VIS) and 752 videos in near-infrared (NIR), spanning across 107 individuals. The dataset incorporates a range of head movements and lighting environments to simulate real-world variability. Video data were acquired using three different cameras: the Logitech C310, the Intel RealSense F200, and the front-facing camera of the HUAWEI P9 smartphone. The dataset employs the CONTEC CMS60C BVP sensor to obtain the ground-truth heart rate measurements.

4.2. Preprocessing

The face of each subject in the video was cropped using the Mediapipe [33] and interpolated to generate a 72 × 72 pixel image. Subsequently, we performed a normalized difference operation [9] between consecutive frames which are defined as follows, $\overline{L}(\lambda,t) = \frac{L(\lambda,t)-L(\lambda,t-1)}{L(\lambda,t)+L(\lambda,t+1)}, t \geq 1$

This aims to reduce the variation of pixel values in different scenes [5]. The frame difference not only provides a temporal context for the underlying network but also reduces the effects of motion artifacts and background pixels. The normalized frame difference was then fed into the rPPG extraction model.

For PPG signals, as the sampling frequencies of PPG signals were different from the video frames per second (FPS), we resampled the PPG signals to the same length as the video frames. Afterward, we detrended the DC components from the PPG signals and then applied a 3rd order bandpass Butterworth filter with frequency between 0.5 Hz and 3 Hz.

The video frames and corresponding PPG signals were then divided into clips of 128 data points in length. This operation resulted in 2800 samples for the BH-rPPG dataset and 3630 samples for the MMPD-rPPG dataset. For the VIPL-HR dataset, we employed the clip length of 160 frames, leading to 7206 samples in total.

4.3. Training configuration

We fine-tuned the image enhancement model using the pre-trained weights provided in the original literature. This fine-tuning process aims to achieve faster convergence and better performance [28]. During training, we employed the Adam optimizer and OneCycleScheduler, where the maximum learning rate was set to $3e^{-4}$, weight decay was set to 0, and momentum was set to 0.9. The number of epochs was 20 and the batch size was 4.

4.4. Loss function

In our previous study [7], we identified a potential discrepancy between the predicted rPPG signal and the ground truth PPG signal as shown in Fig. 4. This inconsistency may stem from issues related to the measurement apparatus. To solve this problem, we proposed the ShiftLoss function for the PhysNet model to reduce the effect of misalignments as shown in Table 2 of [7]. In this work, we generalized the ShiftLoss function to the DeepPhys model by employing ShiftLoss Mean Squared Error (ShiftLoss MSE) as a loss function, which is defined as follows,

$$L_{\mathrm{MSE}} = \min_{\tau\in(-\delta_t,\delta_t)} \sqrt{\sum_t (s(t+\tau)-\hat{s}(t))^2} \tag{1}$$

Where s stands for the predicted rPPG signal and $\hat{s}$ stands for the ground truth PPG signal. The ShiftLoss MSE function can handle the misalignment between the predicted rPPG and ground truth PPG signal by a temporal shift toleration of δ_t. In our experiment, we conducted a statistical analysis of the misalignments and found that the misalignments were Gaussian distributed with a mean of 0 s and a standard deviation of around 1/6 s. Therefore we set δ_t to 1/3 s to ensure that 95% of the misalignments fell within the interval between $[-\delta_t, \delta_t]$.

For the PhysFormer model, we employed a combination of cross-entropy, negative Pearson, and label distribution as loss functions [10] for both PhysFormer training and IEM fine-tuning. The other training parameters were configured according to the same strategy in [10] for better performance.

4.5. Experiments setup

We conducted three experiments. The first experiment was to benchmark the performance of our proposed IEM + rPPG extraction model within different rPPG datasets. The second experiment was to evaluate the inter-dataset performance of our proposed method. The third experiment was to check the efficacy of IEM with the traditional rPPG extraction method.

We employed DeepPhys, PhysNet, and PhysFormer models as the representations of the deep-learning based rPPG extraction model and POS as the representation of traditional method. For the dataset, we selected samples under low illumination conditions in the BH-rPPG dataset and MMPD-rPPG dataset.

4.5.1. Experiment 1: Intra-dataset model performance on low illumination intensity videos

In our first experiment, we employed three setups for each trial:

- Pre-trained DeepPhys/PhysNet/PhysFormer: We pre-trained the rPPG extraction model on the UBFC-rPPG dataset as it contains subjects under normal-light conditions. This setup was used as the baseline.
- IEM + pre-trained DeepPhys/PhysNet/PhysFormer: After pre-training, we froze the parameters of the rPPG extraction model, and cascaded it after the IEM module. Then we trained and tested the framework on the BH-rPPG/MMPD-rPPG/VIPL-HR dataset to evaluate the improved performance of the IEM module by benchmarking with the baseline pre-trained model.
- Re-trained DeepPhys/PhysNet/PhysFormer: One might contend that the improvement of the IEM + pre-trained model comes from the training process in the new low-light dataset (training on BH-rPPG/MMPD-rPPG dataset). To examine the influence of this factor, we further re-trained the DeepPhys/PhysNet/PhysFormer on the two low-light datasets. By comparing the IEM + pre-trained model with the re-trained model, we could exclude the influence of fine-tuning the IEM module in the new low-light dataset.

In this experiment, we trained and tested the model on each dataset individually. To ensure that the training and testing phases did not involve the same trials, we employed five-fold cross-validation, i.e., we split the subjects in each dataset into five groups: four groups were used for training while the fifth group was used for testing to avoid the problem of overlapping data. In the end, we took the average of the five shots as the final result.

4.5.2. Experiment 2: Inter-dataset model performance on low illumination intensity videos

In previous experiments, the training and testing phases were conducted on the same dataset. As a result, over-fitting into the training dataset might happen and the framework would not be able to generalize to other datasets. To test the generalization capability of the proposed IEM + rPPG extraction framework, we designed another two experiments on inter-dataset validation between BH-rPPG dataset and MMPD-rPPG dataset.

In the first experiment, we trained the DeepPhys/PhysNet/ PhysFormer model on the BH-rPPG dataset and tested it on the MMPD-rPPG dataset. In the second experiment, we trained the DeepPhys/PhysNet/ PhysFormer model on the MMPD-rPPG dataset and tested it on the BH-rPPG dataset. The pre-trained models were the same as in the previous experiment.

4.5.3. Experiment 3: evaluation of image enhancement model on traditional method

To evaluate the generalizability of our IEM + rPPG extractor framework outside deep learning methods, we applied it to the POS algorithm, a widely-accepted, analytical method for rPPG extraction. This allowed us to assess whether the benefits of IEM extend to a broader range of rPPG extraction techniques. We benchmarked the performance of the IEM + POS model against a POS-only baseline model using both the BH-rPPG and MMPD-rPPG datasets. Consistent with our previous experiments, we employed five-fold cross-validation during the training of the IEM module.

4.6. Metrics

In this experiment, we calculated heart rate estimation metrics, which were derived based on ground truth PPG signals, including MAE (Mean Absolute Error), RMSE (Root Mean Square Error), MAPE (Mean Absolute Percentage Error), and Pearson's coefficient. The specific definitions of these metrics are given below:

$$MAE_{HR} = \frac{\sum_k |HR_k^{pred} - HR_k^{gt}|}{K}$$

$$RMSE_{HR} = \sqrt{\frac{\sum_k |HR_k^{pred} - HR_k^{gt}|^2}{K}}$$

$$MAPE_{HR} = \frac{1}{K}\sum_{k=1}^{K} \left|\frac{HR_k^{pred} - HR_k^{gt}}{HR_k^{gt}}\right|$$

$$PC_{HR} = \frac{\sum_k (HR_k^{pred} - \overline{HR}^{pred})(HR_k^{gt} - \overline{HR}^{gt})}{\sqrt{\sigma(HR_k^{pred}) \cdot \sigma(HR_k^{gt})}}$$

5. Results

5.1. Exp. 1: Intra-dataset model performance on low illumination intensity videos

5.1.1. Training and testing on BH-rPPG dataset

DeepPhys: The results for the BH-rPPG dataset are shown in Table 1. The pre-trained DeepPhys corresponds to the baseline setup, where the DeepPhys model was pre-trained on the UBFC rPPG dataset without further re-training on the BH-rPPG dataset. As a result of the dataset discrepancy, the baseline pre-trained DeepPhys showed the worst performance among all the setups.

The re-trained DeepPhys corresponds to the setup where the pre-trained DeepPhys model was re-trained on the BH-rPPG dataset which employed the aforementioned five-fold cross-validation strategy. It is noticeable that the MAE, RMSE, and MAPE are less while the Pearson r is greater compared to the baseline, which shows that the re-training on the new dataset can help improve the performance to some extent.

Table 1
Results for HR estimation on BH-rPPG dataset with different setups of DeepPhys, PhysNet, and PhysFormer models. The units of MAE, RMSE and MAPE are beat-per-minute (bpm).

Methods	MAE	RMSE	MAPE	Pearson
Pre-trained DeepPhys[1]	3.817	6.431	5.568	0.850
Re-trained DeepPhys	3.239	5.821	4.622	0.864
IEM + DeepPhys	**2.260**	**3.241**	**3.301**	**0.964**
Pre-trained PhysNet	4.545	9.474	5.726	0.682
Re-trained PhysNet	3.189	6.326	3.955	0.826
IEM + PhysNet	**2.863**	**5.511**	**3.522**	**0.874**
Pre-trained PhysFormer	9.316	13.504	12.336	0.485
Re-trained PhysFormer	2.687	10.179	4.473	0.635
IEM + PhysFormer	**1.708**	**2.122**	**2.251**	**0.985**

IEM + pre-trained DeepPhys is shown in the 3rd row. During the training process, the pre-trained DeepPhys model was frozen and no longer re-trained, and only the IEM was fine-tuned on the BH-rPPG dataset. From Table 1, the IEM + pre-trained DeepPhys achieved the best performance among all the setups. By comparing this with the re-trained DeepPhys, we found out that the improvement in the performance was due not only to the training on the BH-rPPG dataset but also to the employment of the IEM module.

PhysNet: To test the generalization capability of the proposed framework, we replaced the DeepPhys model with the PhysNet model, which is a 3D CNN-based rPPG extractor, and tested its performance with a similar experimental setup. The results are shown in Table 1. It can be seen from Table 1 that the same trend holds for the PhysNet model. The baseline pre-trained PhysNet showed the worst performance and the re-trained PhysNet on the BH-rPPG dataset achieved better performance. Our proposed IEM + PhysNet model performed best among all the setups. However, the results seem different from the results in [7]. This is because of the randomness of grouping the subjects. For example, when the subjects in training and testing groups have more divergent features, the results might not be as good as the scenarios where the subjects in training and testing groups have similar features.

To evaluate the effectiveness of IEM, we conducted Wilcoxon rank test between the results of re-trained PhysNet and IEM + pre-trained PhysNet, and the p-value is 0.058, which is not significant at significance level of 0.05. Nevertheless, it still can be found from the Table 1 that IEM brought more improvement to the rPPG extraction performance under low illumination.

PhysFormer: Table 1 reveals a consistent pattern with the PhysFormer model. Initially, the pre-trained baseline PhysFormer displayed the lowest effectiveness, whereas the PhysFormer re-trained on the BH-rPPG dataset exhibited remarkable enhanced performance and outperformed the previous DeepPhys and PhysNet results, indicating the great fitness ability of PhysFormer. Besides, the combination of our proposed IEM module with the PhysFormer model emerged as the most effective. By comparing the re-trained PhysNet with the IEM-enhanced pre-trained PhysNet, it is evident that the IEM module significantly boosts the rPPG signal extraction capability, especially in conditions of low light.

5.1.2. Training and testing on MMPD-rPPG dataset

To test whether the performance of the improvement on rPPG extraction is an artifact of the specific BH-rPPG dataset, we further train and tested the HR estimation performance of the IEM + rPPG extraction model on MMPD-rPPG dataset under low-light illumination conditions.

DeepPhys: The results for DeepPhys rPPG extraction model on the MMPD-rPPG dataset of low-light conditions are shown in Table 2. It can be seen from the Table that the pre-trained DeepPhys performed worst

Table 2
Results for HR estimation on MMPD-rPPG dataset with different setups of DeepPhys, PhysNet, and PhysFormer models.

Methods	MAE	RMSE	MAPE	Pearson
Pre-trained DeepPhys	6.845	12.140	8.007	0.554
Re-trained DeepPhys	6.179	11.361	7.131	0.606
IEM + DeepPhys	**4.901**	**10.520**	**5.638**	**0.654**
Pre-trained PhysNet	6.525	12.960	9.558	0.446
Re-trained PhysNet	2.797	6.751	3.999	0.834
IEM + PhysNet	**0.799**	**1.515**	**1.007**	**0.993**
Pre-trained PhysFormer	6.490	8.882	6.987	0.533
re-trained PhysFormer	1.256	2.349	1.438	0.972
IEM + PhysFormer	**0.879**	**2.325**	**0.926**	**0.980**

Table 3
Results for HR estimation on VIPL-HR dataset with PhysFormer model.

Methods	MAE	RMSE	MAPE	Pearson
Pre-trained PhysFormer	31.555	35.942	67.265	−0.049
re-trained PhysFormer	10.531	15.532	14.538	0.338
IEM + PhysFormer	**8.352**	**13.819**	**10.514**	**0.515**

(MAE=6.845bpm). After re-training, the performance of the DeepPhys model slightly improved. With the IEM module, the improvement was more significant. The MAE of HR estimation was reduced to less than 5 bpm for IEM + pre-trained DeepPhys.

PhysNet: The results for the PhysNet model on the MMPD dataset are shown in Table 2. In this experiment, the proposed IEM + pre-trained PhysNet achieved the best performance compared to the baseline pre-trained PhysNet and the re-trained PhysNet. It proved that the proposed IEM module could benefit the performance of the deep-learning based rPPG extraction model under low-light conditions regardless of the datasets. The combination of the IEM module and PhysNet model showed great advantage when applied to different categories of subjects, even with different skin tones.

PhysFormer: The outcomes of the PhysFormer model's application to the MMPD dataset are presented in the bottom of Table 2. The combination of the proposed IEM module with the pre-trained PhysFormer outperformed both the standalone pre-trained PhysFormer and the re-trained PhysFormer. This demonstrated the effectiveness of the IEM module in enhancing the performance of deep-learning based rPPG extraction models in low-light environments across various datasets. The integration of the IEM module with the PhysFormer model significantly improved its applicability to diverse subject categories, including those with varying skin tones.

5.1.3. Training and testing on VIPL-HR dataset

PhysFormer: The impact of the IEM method on PhysFormer was further assessed utilizing the VIPL-HR dataset, as indicated in Table 3. Initially, it was evident that PhysFormer, in its pre-trained state, exhibited suboptimal performance on the VIPL-HR dataset, attributed to the dataset's inclusion of diverse, challenging scenarios for rPPG extraction. However, subsequent fine-tuning significantly enhanced PhysFormer's performance. The incorporation of IEM further improved its efficacy, underscoring the effectiveness of IEM in facilitating rPPG extraction across a variety of conditions.

5.2. Experiment 2: Inter-dataset model performance on low illumination intensity videos

5.2.1. Training on BH-rPPG dataset and testing on MMPD-rPPG dataset

In this experiment, we trained the model using all the subjects in the BH-rPPG dataset under low illumination conditions and tested it using all the subjects in the MMPD-rPPG dataset under low light conditions.

DeepPhys: Table 4 delineates the performance metrics of the DeepPhys model when employed as an rPPG signal extractor. The pre-trained DeepPhys model shares the same result as the baseline setup in Table 2. Subsequent re-training of the DeepPhys model yielded only marginal improvements in performance metrics. However, the integration of the proposed IEM framework with the pre-trained DeepPhys model resulted in a notable reduction in the mean absolute error (MAE) between the actual HR and the HR estimated by the model, diminishing it to under 4 bpm. This enhancement underscores the proficiency of the proposed IEM-enhanced DeepPhys model in generalizing across different datasets, specifically from the BH-rPPG dataset to the MMPD-rPPG dataset.

Table 4
Results for HR estimation of the DeepPhys, PhysNet, and PhysFormer model for training on BH-rPPG dataset and testing on MMPD-rPPG .dataset.

Methods	MAE	RMSE	MAPE	Pearson
Pre-trained DeepPhys	6.845	12.140	8.007	0.554
Re-trained DeepPhys	5.273	10.910	6.125	0.629
IEM + DeepPhys	**3.596**	**7.377**	**4.229**	**0.826**
Pre-trained PhysNet	6.525	12.960	9.558	0.446
Re-trained PhysNet	29.590	37.980	43.560	0.279
IEM + PhysNet	**6.099**	**12.38**	**9.851**	**0.431**
Pre-trained PhysFormer	6.490	8.882	6.987	0.533
Re-trained PhysFormer	7.032	9.317	7.185	0.498
IEM + PhysFormer	**5.315**	**6.490**	**5.547**	**0.668**

Table 5
Results for HR estimation of DeepPhys, PhysNet, and PhysFormer models for training on MMPD-rPPG dataset and testing on BH-rPPG dataset.

Methods	MAE	RMSE	MAPE	Pearson
Pre-trained DeepPhys	3.817	6.431	5.568	0.850
Re-trained DeepPhys	4.244	7.969	5.837	0.752
IEM + DeepPhys	**3.544**	**6.019**	**4.158**	**0.878**
Pre-trained PhysNet	**4.545**	**9.474**	**5.726**	**0.682**
Re-trained PhysNet	5.399	12.129	7.757	0.556
IEM + PhysNet	5.12	10.442	6.382	0.579
Pre-trained PhysFormer	9.316	13.504	12.336	0.485
Re-trained PhysFormer	10.645	13.790	11.084	0.320
IEM + PhysFormer	**8.958**	**10.340**	**9.453**	**0.529**

PhysNet: The results for PhysNet of training on the BH-rPPG dataset and testing on the MMPD-rPPG dataset can be found in Table 4. In contrast to the DeepPhys model, PhysNet exhibited markedly poor performance under the same dataset configuration. This performance discrepancy can be attributed to the overfitting of PhysNet on the training dataset. The incorporation of the IEM module into the PhysNet framework ameliorated this issue to some extent, reducing the estimation error of the pre-trained PhysNet model.

PhysFormer: The results for PhysFormer of training on the BH-rPPG dataset and testing on the MMPD-rPPG dataset can be found at the bottom of Table 4. The re-training on the BH-rPPG dataset slightly degrades the performance of PhysFormer on the MMPD-rPPG dataset. Despite the notable enhancement observed with the IEM + pre-trained model on the BH-rPPG dataset, fine-tuning the IEM module did not markedly affect its performance as much as re-training the PhysFormer model did. Even so, the performance was improved in the inter-dataset test on the MMPD-rPPG dataset with the help of IEM. This result proves the generalization capacity of the IEM module to unseen examples.

5.2.2. Training on MMPD-rPPG dataset and testing on BH-rPPG dataset

In opposite to before, we trained the model using all the subjects in the MMPD-rPPG dataset under low illumination conditions and tested it using all the subjects in the BH-rPPG dataset under low light conditions in this experiment.

DeepPhys: The results for the DeepPhys model as an rPPG extractor are shown in Table 5. For the re-trained DeepPhys model, the performance was degraded, which indicated that the re-training process would inevitably influence the generalization capacity. On the other front, the employment of the IEM still improved the HR estimation

Table 6
HR estimation of POS algorithm on BH-rPPG dataset and MMPD-rPPG dataset. The dataset can be found in square brackets in the first column.

Methods	MAE	RMSE	MAPE	Pearson
POS[BH→BH]	6.454	11.170	8.290	0.563
IEM + POS[BH→BH]	**3.390**	**6.606**	**4.345**	**0.819**
POS[MMPD→MMPD]	12.278	18.105	14.257	0.120
IEM + POS[MMPD→MMPD]	**8.020**	**11.122**	**10.423**	**0.505**
POS[BH→MMPD]	12.278	18.105	14.257	0.120
IEM + POS[BH→MMPD]	**10.890**	**14.438**	**11.129**	**0.388**
POS[MMPD→BH]	6.454	11.170	8.290	0.563
IEM + POS[MMPD→BH]	**5.588**	**8.103**	**6.479**	**0.605**

accuracy. This demonstrated that IEM was well-suited to adapting across various datasets.

PhysNet: The results for PhysNet of training on the MMPD-rPPG dataset and testing on the BH-rPPG dataset can be found in Table 5. Unlike the DeepPhys model, the pre-trained PhysNet model achieved the best performance while both the re-trained PhysNet and IEM + pre-trained model showed worse performance. This means the re-training and fine-tuning of the MMPD-rPPG dataset would unavoidably affect the performance on the BH-rPPG dataset, which is in contrast to the results in previous experiments.

PhysFormer: From the data presented in Table 5, it is evident that re-training adversely affected the performance of PhysFormer when it was trained on the MMPD-rPPG dataset and tested on the BH-rPPG dataset. In contrast to the results with PhysNet, the incorporation of the IEM module with PhysFormer led to enhanced performance relative to the baseline pre-trained PhysFormer. This improvement underscores the effectiveness of the IEM module in augmenting rPPG signal extraction capabilities in low-light conditions.

5.3. Exp. 3: Evaluation of the image enhancement model on traditional method

The results for comparison of the baseline POS algorithm with the IEM + POS algorithm on both the BH-rPPG dataset and the MMPD-rPPG dataset are shown in Table 6. As shown, the HR estimation accuracy was improved in both datasets with the employment of the IEM module compared to using POS only for intra-dataset and inter-dataset cases. This result confirms that our proposed framework can also be generalized to analytical traditional methods. We also found that HR estimation error was larger on the MMPD-rPPG dataset compared to the BH-rPPG dataset as a result of the diversity of the scenarios in the MMPD-rPPG dataset, and the POS algorithm underwent the influence of different color spaces forced by features such as skin tone.

6. Discussion

We summarized the HR MAE results of all the experiments in Fig. 5. Besides, we also demonstrated the HR estimation curve using Bland-Altman Plots in Fig. 6. Across most the experiments, the IEM significantly enhanced the models' HR estimation accuracy in low-light conditions. The performance gain were more pronounced in scenarios where the data used for testing and training were from the same source (intra-dataset), compared to those where the data came from different sources (inter-dataset). By conducting Wilcoxon signed rank test, the p-values are less than 0.05 for experiments where the models were trained and tested on the dataset, showing that the IEM can significantly reduce the estimation error of HR. For inter-dataset scenarios, there are still slight reduction of HR estimation error but not as significant as intra-dataset scenarios. It should be noted the only exception of PhysNet model training on the MMPD-rPPG dataset and testing on the BH rPPG dataset, where the employment of IEM degraded the performance. We ascribe this phenomenon to the diversity of the subject condition in the MMPD-rPPG dataset employed in this experiment. According to the previous study [34], different features, including gender, and skin tone, will influence the results of rPPG extraction and lead to bias in the HR estimation.

Table 7
Analysis of variance (ANOVA) on different features: skin tone, glasses wearing, hair covering and makeup, in MMPD dataset for HR estimation absolute error with IEM + pre-trained PhysNet model. Items are in ascending order according to the p-values.

Features	Options	F stats	p-value
Skin Tone	3/4, 5, 6	5.758	**0.023**
Glasses Wearing	True/False	1.364	0.252
Hair Covering	True/False	1.033	0.317
Makeup	True/False	0.695	0.411

Table 8
HR estimation of PhysNet model for training on MMPD-rPPG dataset skin tone 3 and testing on BH-rPPG dataset.

Methods	MAE	RMSE	MAPE	Pearson
Pre-trained PhysNet	4.545	9.474	5.726	0.682
Re-trained PhysNet	5.499	12.150	7.878	0.561
IEM + PhysNet	**3.663**	**7.202**	**4.972**	**0.829**

To find out the potential cause of this phenomenon, we further checked the HR estimation error of each trial. Analysis of variance (ANOVA) for absolute HR estimation error between the predicted HR and ground truth HR on each feature (i.e. skin tone, makeup, glasses wearing, hair covering) was conducted. For skin tone factor, we divided the subjects into two groups. One group contained subjects with skin tone 3 (Fitzpatrick skin type 3) and the other contained subjects with skin tones 4, 5, and 6 (Fitzpatrick skin types 4, 5, and 6). This is attributed to the majority of subjects in the BH-rPPG dataset having skin tone 3. The corresponding F statistics and p-values of each feature are shown in Table 7.

As can be seen from the Table, skin tone 3 showed a significant effect at 0.05 level. We found that skin tone 3 was similar to the subjects in the BH-rPPG dataset. By checking the corresponding errors on each skin tone, we found that subjects corresponding to skin tone 3 had the smallest HR estimation error. The statistical fact indicated that skin tone was the potential cause for performance degradation as skin color would influence the training of IEM for image decomposition.

Therefore, we re-conducted this experiment. Instead of training on all the subjects in MMPD, we selected those subjects with skin tone 3 only. The results are shown in Table 8. We observed that the IEM + pre-trained model achieved the best performance among the models. The result confirms the conclusion that skin tone 3 is the factor that significantly influences the performance of HR estimation, indicating that the distribution of various factors within the dataset has the potential to exert influence.

In real-world applications, the challenge of different skin tones is unavoidable. To eliminate the effect of skin tone, a potential strategy is skin tone normalization. This involves converting the RGB image to the YCrCb color space and normalizing it against a reference skin tone standard. This approach remains a speculative hypothesis and falls outside the current study's scope. Future research will be dedicated to exploring and validating solutions to address this challenge of skin tone.

7. Conclusions

In this paper, we propose an IEM + deep-learning based rPPG extractor to improve the extracted rPPG signal for HR estimation accuracy in low light conditions. Our method utilizes transfer learning to fine-tune the IEM to be able to collect rPPG-related signals from raw image sequence input. We also propose an improved ShiftLoss function that can be adapted to more scenarios to handle the misalignment between the video and ground truth PPG label. Our results demonstrate that the

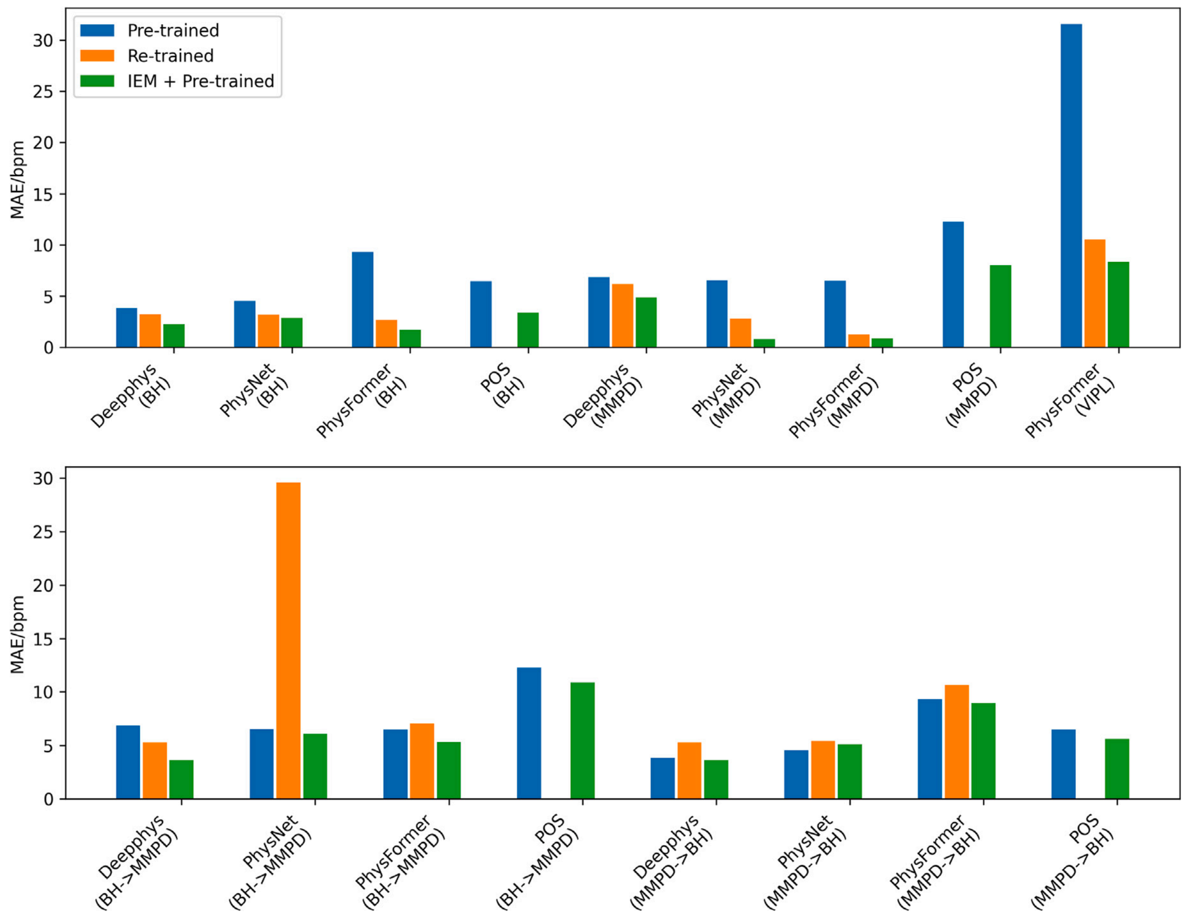

Fig. 5. HR MAE of all the experiments for different methods and different datasets. The top figure illustrates the intra-dataset cases and the bottom figure illustrates the inter-dataset cases. On the left and right sides of the arrows in the brackets on the x-axis are the training dataset and testing dataset, respectively (For example, BH->MMPD means training on BH-rPPG dataset and testing on MMPD-rPPG dataset.

employment of IEM would improve the quality of the rPPG signal under low illumination levels. By employing different representative deep-learning based rPPG extractors (DeepPhys, PhysNet, and PhysFormer) and a traditional method (POS), we showed the great generalization capability of the proposed IEM + rPPG extractor framework. We also conducted inter-dataset validation, which showed that our proposed IEM + rPPG extractor framework is generally applicable to different datasets. However, different factors like skin tone would influence the generalization capacity.

Future work will be concentrated on getting over the influence of different subject appearance factors, such as skin tone, gender, makeup, glasses, etc. Experiments with more diverse setups including a greater variety of deep-learning based rPPG extraction models to produce a more general IEM model that works for different rPPG extractors will be done. More datasets should also be evaluated including dynamic illumination conditions in future work. Nevertheless, the consistent improvements demonstrate the general validity of the proposed image enhancement-based method.

Funding

This paper is partially supported by the Innovation and Technology Commission via the TSSSU Fund and internship program. The authors are also grateful for the support of the HKSTP incubation program.

CRediT authorship contribution statement

Shutao Chen: Writing – original draft, Software, Methodology. **Kwan Long Wong:** Resources, Formal analysis. **Tze Tai Chan:** Supervision, Methodology. **Ya Wang:** Methodology, Formal analysis. **Richard Hau Yue So:** Supervision, Formal analysis. **Jing Wei Chin:** Supervision, Software.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Shutao CHEN reports financial support was provided by Innovation and Technology Commission. Shutao CHEN reports a relationship with Innovation and Technology Commission that includes: funding grants. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. Shutao Chen, Kwan Long Wong, Jing Wei Chen, and Tsz Tai Chan report financial support was provided by PanopticAI for this research. While this study explores the use of POS and other deep learning models, this should not be interpreted as an endorsement or adoption of these specific algorithms by PanopticAI. PanopticAI's commercial software employs proprietary solutions that are distinct from the methods investigated in this paper and do not infringe on any commercial rights. All other authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

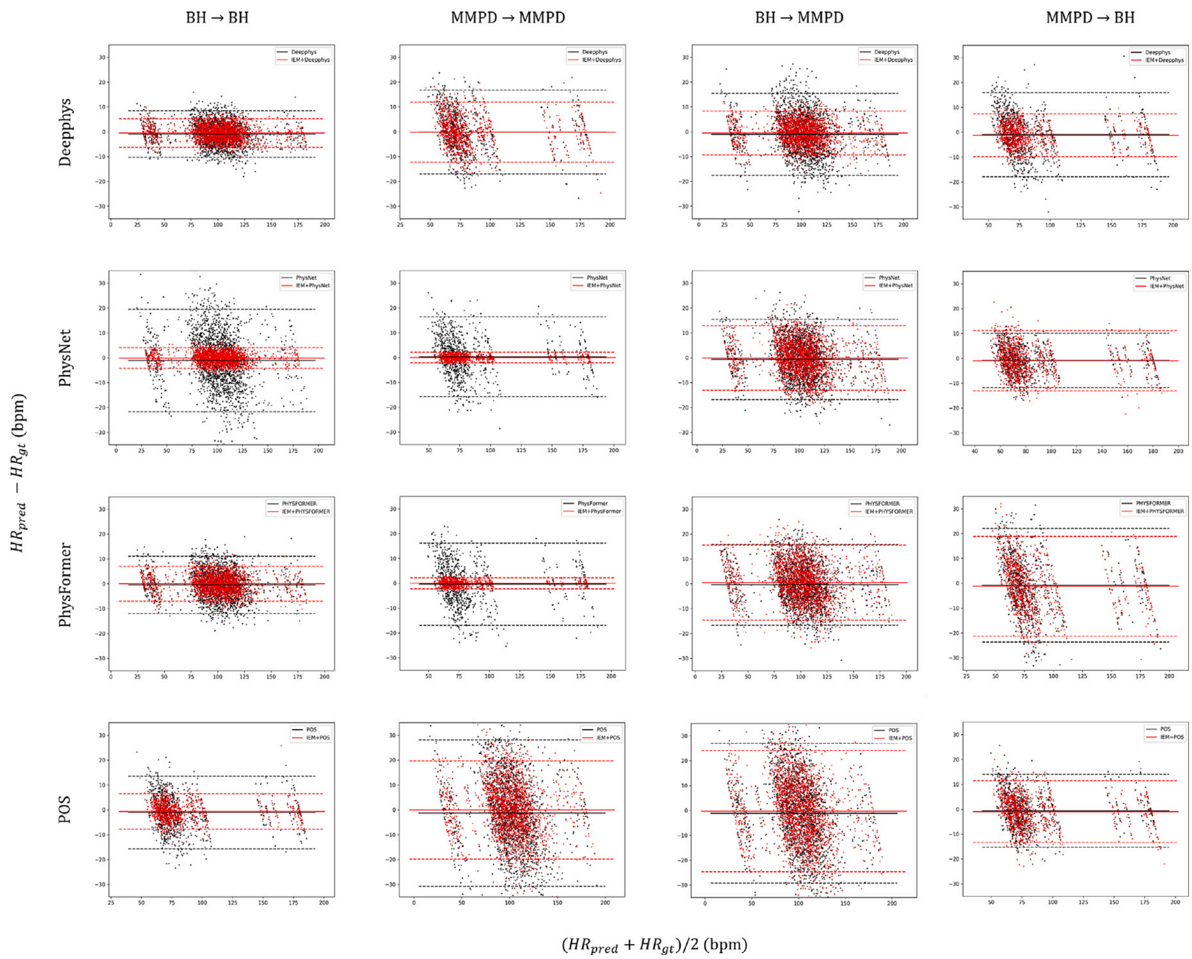

Fig. 6. Bland-Altman Plots of all the MAE of HR estimation for different methods on all the intra-dataset and inter-dataset cases. Black and red dots correspond to the results before and after processing by the IEM module. On the left and right sides of the arrows in the brackets on the x-axis are the training dataset and testing dataset, respectively (For example, BH->MMPD means training on BH-rPPG dataset and testing on MMPD-rPPG dataset.

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